

STATSIG lecture 5

Basic discrete-time Kalman filter (KF)

for linear systems

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Discrete-time Kalman filter

State vector x_k , Input vector u_k , Process noise w_k , Measurement noise v_k

State Space model process model

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}$$

zero-mean process noise covariance matrix $Q_k = \text{Cov}[w_k]$

observation model

$$z_k = Hx_k + v_k$$

zero-mean measurement noise covariance matrix $R_k = \text{Cov}[v_k]$

Notation

$\hat{x}_k^-$ = a priori state estimate (before measuring z_k)

$\hat{x}_k$ = posterior state estimate (after measuring z_k)

$\hat{z}_k^-$ = a priori estimate of next measurement

$P_k^- = E[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T]$ (a priori MSE estimate)

$P_k = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T]$ (posterior MSE estimate)

Time line from sample time $k-1 \rightarrow k$

$k-1$

k

Predict step $\rightarrow$ Update step

Predict step

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_{k-1}$$

$$\hat{z}_k^- = H\hat{x}_k^-$$

$$P_k^- = AP_{k-1}A^T + Q_{k-1}$$

Update step

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - \hat{z}_k^-)$$

where Kalman Gain (minimizing MSE)

$$K_k = P_k^- H^T (H P_k^- H^T + R_k)^{-1}$$

$$P_k = (I - K_k H) P_k^-$$

$$k \rightarrow k + 1$$

Algorithm

initialize, e.g. $\hat{x}_0 = 0$, $P_0 = I$, $k = 1$

iterate: predict, update, predict, update